

Q3: An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach..

Introduction

Aliasing is a distortion that occurs in digital images when the sampling process is not performed correctly. It appears as jagged edges, false patterns (moiré), or loss of fine details. Sampling and quantization are two important steps in digital image processing that directly affect image quality.

Cause of Aliasing

1. Improper Sampling

Sampling is the process of converting a continuous image into a digital image by representing it with pixels.

- If the **sampling rate is too low**, the image sensor cannot capture all the fine details of the original scene.
- High-frequency details are incorrectly represented, producing **aliasing artifacts**.
- This happens because the sampling frequency is lower than the **Nyquist rate**, which states that the sampling frequency should be at least twice the highest frequency present in the image.

2. Effect of Quantization

Quantization is the process of assigning discrete intensity values to each pixel.

- If the number of quantization levels is too low (low bit depth), the image loses smooth intensity variations.
- This results in **banding**, false intensity transitions, and reduced image quality.
- Although quantization mainly affects brightness and gray levels, insufficient quantization can make aliasing effects more noticeable.

Effects of Aliasing

Aliasing produces several visible defects in the image, such as:

- Jagged or staircase edges.
- Moiré patterns.
- Loss of fine details.
- Distorted textures.
- Reduced overall image quality.

Corrective Approaches

The aliasing problem can be minimized using the following methods:

1. Increase the Sampling Rate

- Capture the image at a higher resolution using more pixels.
- A higher sampling rate preserves fine details and reduces aliasing.

2. Apply an Anti-Aliasing Filter

- Use a low-pass or Gaussian filter before sampling.
- This removes high-frequency components that cause aliasing.

3. Use Better Interpolation Techniques

- When resizing images, use bilinear or bicubic interpolation instead of nearest-neighbor interpolation.
- These methods produce smoother edges and reduce distortions.

4. Increase Quantization Levels

- Use a higher bit depth (such as 8-bit or 16-bit).
- This provides smoother intensity transitions and improves overall image quality.

Conclusion

Aliasing occurs mainly because of **insufficient sampling**, while **low quantization levels** reduce the accuracy of intensity representation. The problem can be minimized by increasing the sampling rate, applying anti-aliasing filters, using better interpolation methods, and increasing the number of quantization levels. These techniques improve image quality and produce a more accurate digital representation of the original scene.